# 6710503739_HW#8_Exercise-1

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## 0.1 Find all frequent itemsets

### 0.1.1 Candidate 1-itemsets L1

| Itemset | Count | Support | Result |
|---|---|---|---|
| {F} | 1 | 1/6 = 16.67% | no |
| {G} | 6 | 6/6 = 100% | YES |
| {H} | 3 | 3/6 = 50% | YES |
| {I} | 4 | 4/6 = 66.67% | YES |
| {J} | 6 | 6/6 = 100% | YES |

L1 = { {G}, {H}, {I}, {J} }

### 0.1.2 Candidate 2-itemsets L2

L2 is formed by joining L1 with itself

| Itemset | Count | Support | Result |
|---|---|---|---|
| {G,H} | 3 | 3/6 = 50% | YES |
| {G,I} | 4 | 4/6 = 66.67% | YES |
| {G,J} | 6 | 6/6 = 100% | YES |
| {H,I} | 1 | 1/6 = 16.67% | no |
| {H,J} | 3 | 3/6 = 50% | YES |
| {I,J} | 4 | 4/6 = 66.67% | YES |

L2 = { {G,H}, {G,I}, {G,J}, {H,J}, {I,J} }

### 0.1.3 Candidate 3-itemsets L3

L3 is formed by joining L2 with itself

| Candidate | 2-subsets | Action |
|---|---|---|
| {G,H,I} | {G,H} YES, {G,I} YES, {H,I} no | pruned |
| {G,H,J} | {G,H} YES, {G,J} YES, {H,J} YES | count |
| {G,I,J} | {G,I} YES, {G,J} YES, {I,J} YES | count |
| {H,I,J} | {H,I} no, {H,J} YES, {I,J} YES | pruned |

Counting the surviving candidates:

| Itemset | Count | Support | Result |
|---|---|---|---|
| {G,H,J} | 3 | 3/6 = 50% | YES |
| {G,I,J} | 4 | 4/6 = 66.67% | YES |

L3 = { {G,H,J}, {G,I,J} }

#### 0.1.4 Candidate 4-itemsets L4

Joining L3 gives the single candidate {G,H,I,J}. Its 3-subsets are {G,H,I}, {G,H,J}, {G,I,J}, {H,I,J}. Since {G,H,I} not in L3 and {H,I,J} not in L3, the candidate is pruned.

L4 = { } (empty) → the algorithm stops.

#### 0.1.5 Answer 1 — All frequent itemsets

| # | Frequent itemset | Count | Support |
|---|---|---|---|
| 1 | {G} | 6 | 100.00% |
| 2 | {H} | 3 | 50.00% |
| 3 | {I} | 4 | 66.67% |
| 4 | {J} | 6 | 100.00% |
| 5 | {G, H} | 3 | 50.00% |
| 6 | {G, I} | 4 | 66.67% |
| 7 | {G, J} | 6 | 100.00% |
| 8 | {H, J} | 3 | 50.00% |
| 9 | {I, J} | 4 | 66.67% |
| 10 | {G, H, J} | 3 | 50.00% |
| 11 | {G, I, J} | 4 | 66.67% |

Total = 11 frequent itemsets. The maximal frequent itemsets are {G,H,J} and {G,I,J}.

### 0.2 Three strong association rules

#### 0.2.1 1 {H} => {G, J}

Generated from the frequent itemset {G,H,J} (support 3/6).

- support = support({G,H,J}) = 3/6 = 0.5
- confidence = support({G,H,J}) / support({H}) = (3/6) / (3/6) = 3/3 = 1 >= 75% -> strong
- lift = confidence / support({G,J}) = 1 / 1 = 1

#### 0.2.2 2 {G, I} => {J}

Generated from the frequent itemset {G,I,J} (support 4/6).

- support = support({G,I,J}) = 4/6 = 0.667
- confidence = support({G,I,J}) / support({G,I}) = (4/6) / (4/6) = 4/4 = 1 >= 75% -> strong
- lift = confidence / support({J}) = 1 / 1 = 1

#### 0.2.3 3 {I} => {G}

Generated from the frequent itemset {G,I} (support 4/6).

- support = support({G,I}) = 4/6 = 0.667
- confidence = support({G,I}) / support({I}) = (4/6) / (4/6) = 4/4 = 1 >= 75% -> strong
- lift = confidence / support({G}) = 1 / 1 = 1